

HARIKRISHNAN H

AI/ML Engineer | M.Tech Artificial Intelligence | AI Security & Adversarial ML

Kerala, India

☎ +91 7902485392 ✉ hh7662202@gmail.com 🔗 linkedin.com/in/harikrishnan-h911 🌐 github.com/midnightcoder04

National AI Buildathon winner (2026) | M.Tech AI candidate at CET Trivandrum specialising in adversarial attacks on LLMs, deep learning robustness, and production AI pipelines. Targeting AI/ML research or engineering internships.

Education

College of Engineering Trivandrum

Master of Technology in Artificial Intelligence

2025 – Present

Trivandrum, Kerala

- Research Focus: Adversarial Attacks on AI & ML - robustness of deep learning models and LLMs against adversarial perturbations.

College of Engineering Perumon

Bachelor of Technology in Computer Science | CGPA: 8.15 / 10

2021 – 2025

Kollam, Kerala

Achievements

1st Prize (Student Category) | India AI Impact Buildathon 2026 [cert]

2026

- Engineered a multilingual **AI voice fraud detection API** trained on ~120K samples (Mozilla Common Voice, ElevenLabs, Kaggle), achieving **95%+ validation accuracy** at **4ms CPU inference per 5-second clip** on Apple M4 — no GPU required.

First Prize Winner, Hack with CEP | 24-Hour Innovation Hackathon [cert]

2023

Research

Black-Box Adversarial Attack Framework for LLMs

2025 – Present

MTech Research Project | Python, HuggingFace Transformers, Prompt Engineering

CET Trivandrum

- Designed a **text-based black-box adversarial attack framework** across 4 attack families (Role-Playing, Linguistic Obfuscation, Logical Exploits, Multi-Turn) targeting 5 open-source LLMs without access to model weights or gradients.
- Evaluated on a controlled 5-model benchmark (115 trials, 23 prompts/model); achieved overall ASR of **76.9%–100%**. Multi-Turn attacks reached **96% ASR** (25/26 trials).
- Surfaced alignment gaps: Linguistic Obfuscation succeeded only on Vicuna and Mistral (40% overall), exposing model-family inconsistencies. Manuscript in preparation; results presented internally at CET Trivandrum.

Experience & Projects

Chaudary & Associates [legal-action.in]

Apr 2026

Automation Engineer | Python, Tesseract OCR, Zoho ZeptoMail, REST API

- End-to-end email automation pipeline processing ~**5,000 legal PDFs** via Tesseract OCR; SMTP dispatch via Zoho ZeptoMail API saved **80 hrs** of manual work and reduced bounce rate from 3% to 1.8%.

Surgical Scene Inpainting via Attention GAN [github]

Mar 2026

Independent Research | PyTorch, GAN, U-Net, VGG19, SSIM, transfer learning

- Built an attention-enhanced Pix2Pix GAN on the **ISBI 2024 DREAMING dataset** to reconstruct surgical regions occluded by instruments/hands; upgraded standard U-Net with **custom attention blocks on skip connections** for selective feature focus.
- Optimised composite loss (GAN + L1 + SSIM + VGG19 Perceptual); achieved **SSIM 0.9422** and **PSNR 31.46 dB** — competitive with ISBI 2024 baselines (ResNet50/VGG16 encoders, zero-shot ProPainter).

Satellite Image Super-Resolution — GeoUpscale [github]

Dec 2025

Independent Project | PyTorch, ESRGAN, CUDA, Streamlit

- Trained **ESRGAN** (16 RRDB blocks, 64 filters) on SpaceNet imagery for **4× super-resolution** (64×64 → 256×256); composite loss (L1 + VGG19 Perceptual + Adversarial) over 200 epochs with automatic CUDA/CPU detection.
- Achieved **PSNR 24.58 dB** and **SSIM 0.6324** on Kaggle 2×T4 GPUs under significant hardware constraints (64×64 input scale); deployed as a **Streamlit web app** with drag-and-drop upload and real-time preview.

Smart Parking Detection System [github]

Nov 2025

Independent Project | PyTorch, CNN, OpenCV

- Trained a custom CNN on 6,090 labelled images (balanced empty/occupied split), achieving **100% test accuracy** on a held-out 20% split; model generalised to human-occupied spaces despite training exclusively on cars, validating learned spatial occlusion features.

AI Voice Interviewer [github]

Mar 2025

Independent Project | LLM, RAG, LangChain, React, Flask, REST API

- Built a voice-interactive interview platform using locally hosted **DeepSeek-R1 (7B)** via Ollama with real-time STT/TTS pipelines (React + Flask REST API); RAG-style resume parsing via **LangChain** generates context-aware questions and structured scoring.

Technical Skills

Languages: Python, JavaScript, TypeScript, C, SQL

AI & ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, HuggingFace Transformers, CUDA, transfer learning, fine-tuning, Kaggle

LLM Systems: LangChain, Ollama, local model inference, RAG pipelines, voice I/O (STT/TTS)

Web & Infrastructure: React, Next.js, Flask, REST API, Tailwind CSS, Docker, Git, Linux

Certifications

NPTEL Elite — Data Science for Engineers [cert] (IIT Madras, Jan–Mar 2026). Cert. No: NPTEL26CS65S460200259.